



```
In [1]: import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: income_df=pd.read_csv(r'C:\Users\HP\Downloads\Inc_Exp_Data.csv')
```

```
In [3]: income_df.head(2)
```

```
Out[3]:
```

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt
0	5000	8000	3	2000
1	6000	7000	2	3000

```
In [4]: income_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Mthly_HH_Income                       50 non-null    int64
1   Mthly_HH_Expense                       50 non-null    int64
2   No_of_Fly_Members                     50 non-null    int64
3   Emi_or_Rent_Amt                       50 non-null    int64
4   Annual_HH_Income                      50 non-null    int64
5   Highest_Qualified_Member              50 non-null    object
6   No_of_Earning_Members                 50 non-null    int64
dtypes: int64(6), object(1)
memory usage: 2.9+ KB
```

```
In [30]: income_df.shape
```

```
Out[30]: (50, 7)
```

```
In [5]: income_df.describe()
```

```
Out[5]:
```

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_A
count	50.000000	50.000000	50.000000	50.000000
mean	41558.000000	18818.000000	4.060000	3060.000000
std	26097.908979	12090.216824	1.517382	6241.434900
min	5000.000000	2000.000000	1.000000	0.000000
25%	23550.000000	10000.000000	3.000000	0.000000
50%	35000.000000	15500.000000	4.000000	0.000000
75%	50375.000000	25000.000000	5.000000	3500.000000
max	100000.000000	50000.000000	7.000000	35000.000000

```
In [6]: income_df.describe().T
```

```
Out[6]:
```

	count	mean	std	min	25%	75%	max
Mthly_HH_Income	50.0	41558.00	26097.908979	5000.0	23550.0	35000.0	100000.0
Mthly_HH_Expense	50.0	18818.00	12090.216824	2000.0	10000.0	15000.0	50000.0
No_of_Fly_Members	50.0	4.06	1.517382	1.0	3.0	4.0	10.0
Emi_or_Rent_Amt	50.0	3060.00	6241.434948	0.0	0.0	10000.0	50000.0
Annual_HH_Income	50.0	490019.04	320135.792123	64200.0	258750.0	440000.0	1000000.0
No_of_Earning_Members	50.0	1.46	0.734291	1.0	1.0	2.0	5.0

```
In [7]: income_df.isna().any()
```

```
Out[7]: Mthly_HH_Income      False
Mthly_HH_Expense      False
No_of_Fly_Members      False
Emi_or_Rent_Amt        False
Annual_HH_Income      False
Highest_Qualified_Member False
No_of_Earning_Members  False
dtype: bool
```

```
In [8]: income_df.columns
```

```
Out[8]: Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',
               'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',
               'No_of_Earning_Members'],
              dtype='object')
```

```
In [9]: income_df.columns
```

```
Out[9]: Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',
               'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',
               'No_of_Earning_Members'],
              dtype='object')
```

```
In [31]: income_df["Mthly_HH_Expense"].mean()
```

```
Out[31]: np.float64(18818.0)
```

```
In [32]: income_df["Mthly_HH_Expense"].median()
```

```
Out[32]: 15500.0
```

```
In [10]: print(income_df['Mthly_HH_Expense'].mean())
```

```
18818.0
```

```
In [11]: print(income_df['Mthly_HH_Expense'].median())
```

15500.0

```
In [12]: print(income_df['Mthly_HH_Expense'].mode())
```

0 25000

Name: Mthly_HH_Expense, dtype: int64

```
In [13]: mth_exp_tmp = pd.crosstab(index=income_df["Mthly_HH_Expense"], columns="count")
mth_exp_tmp.reset_index(inplace=True)
mth_exp_tmp[mth_exp_tmp['count']] == income_df.Mthly_HH_Expense.value_counts().
```

```
Out[13]:
```

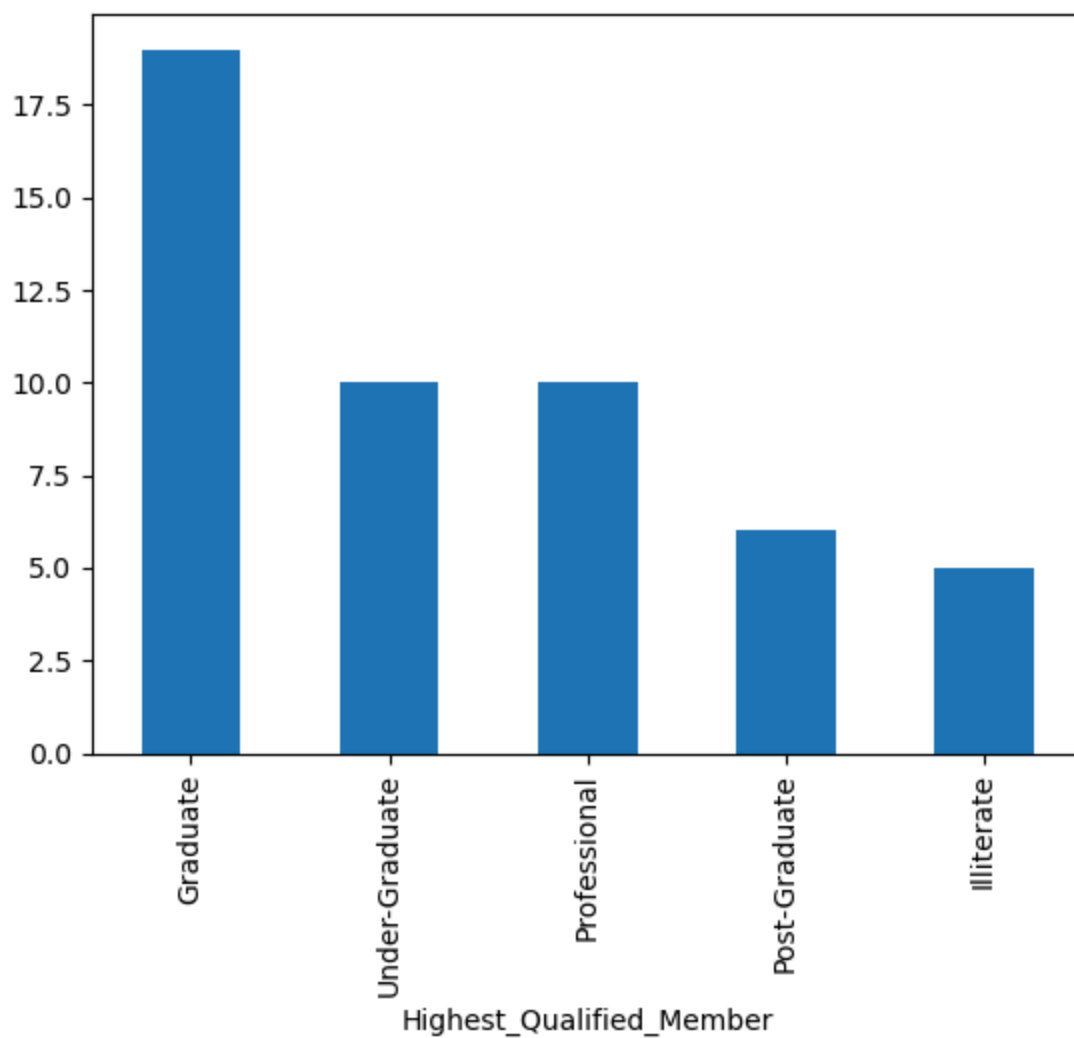
col_0	Mthly_HH_Expense	count
18	25000	8

```
In [14]: income_df.columns
```

```
Out[14]: Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',
               'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',
               'No_of_Earning_Members'],
              dtype='object')
```

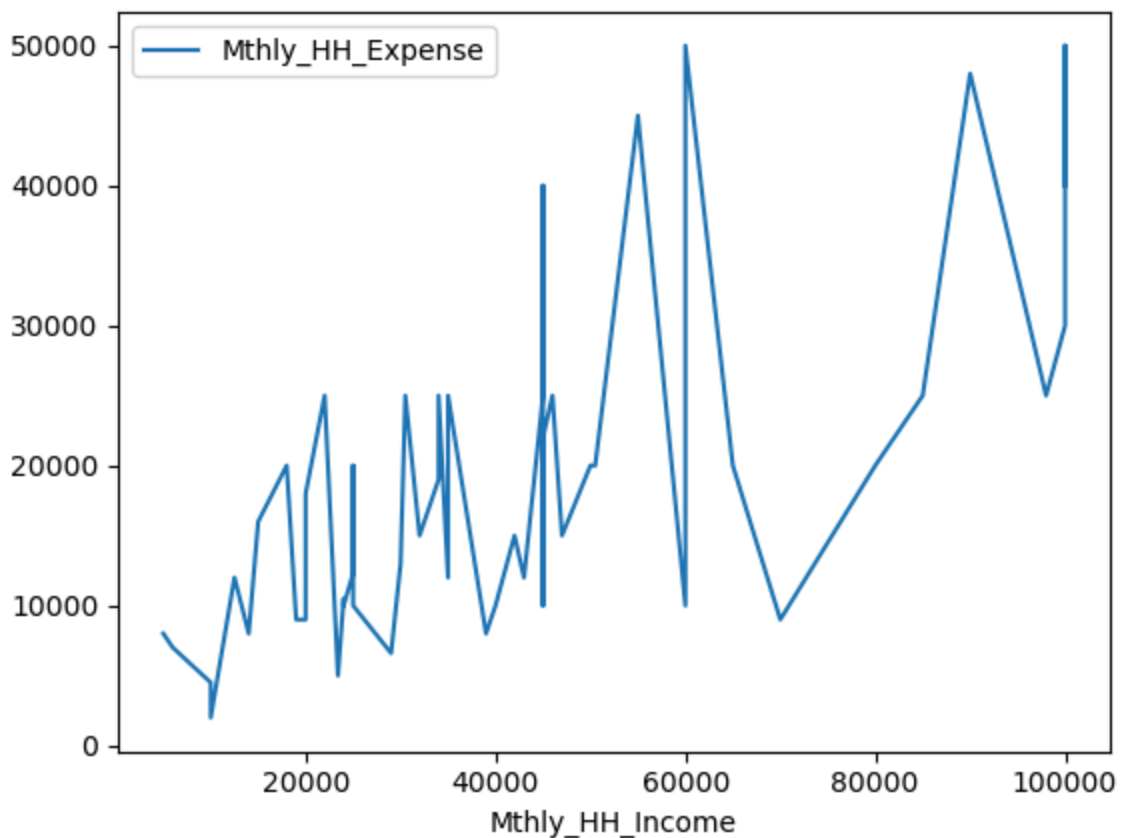
```
In [15]: income_df["Highest_Qualified_Member"].value_counts().plot(kind="bar")
```

```
Out[15]: <Axes: xlabel='Highest_Qualified_Member'>
```



```
In [16]: income_df.plot(x="Mthly_HH_Income", y="Mthly_HH_Expense")
```

```
Out[16]: <Axes: xlabel='Mthly_HH_Income'>
```



```
In [33]: pd.DataFrame(income_df.iloc[:,0:5].std().to_frame()).T
```

```
Out[33]:
```

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt
0	26097.908979	12090.216824	1.517382	6241.434948

```
In [34]: pd.DataFrame(income_df.iloc[:,0:4].var().to_frame()).T
```

```
Out[34]:
```

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt
0	6.811009e+08	1.461733e+08	2.302449	3.895551e+07

```
In [35]: income_df["Highest_Qualified_Member"].value_counts().to_frame().T
```

```
Out[35]:
```

Highest_Qualified_Member	Graduate	Under-Graduate	Professional	Post-Graduate	Illiterat
count	19	10	10	6	

```
In [37]: income_df["Highest_Qualified_Member"].value_counts().to_frame().T
```

```
Out[37]:
```

Highest_Qualified_Member	Graduate	Under-Graduate	Professional	Post-Graduate	Illiterat
count	19	10	10	6	

In [38]: *#Here we need to calculate the coeff of variation*

```
Coeff_of_var_StockA=10/15  
print(Coeff_of_var_StockA)  
Coeff_of_var_StockB=5/10  
print(Coeff_of_var_StockB)
```

```
0.6666666666666666  
0.5
```

In [17]: `print(income_df["Mthly_HH_Expense"].quantile(0.75))`

```
25000.0
```

In [18]: `print(income_df["Mthly_HH_Expense"].quantile(0.50))`

```
15500.0
```

In [19]: `print(income_df["Mthly_HH_Expense"].quantile(0.25))`

```
10000.0
```

In [20]: `IQR=income_df["Mthly_HH_Expense"].quantile(0.75)-income_df["Mthly_HH_Expense"]`

In [21]: `IQR`

Out[21]: `np.float64(15000.0)`

In [22]: `IQR=income_df["Mthly_HH_Expense"].quantile(0.50)-income_df["Mthly_HH_Expense"]`

In [23]: `IQR`

Out[23]: `np.float64(5500.0)`

In [24]: `income_df['Mthly_HH_Expense'].var()`

Out[24]: 146173342.85714287

In [25]: `income_df['Mthly_HH_Expense'].std()`

Out[25]: 12090.216824240286

In [26]: `coef = income_df['Mthly_HH_Expense'].std() / income_df['Mthly_HH_Expense'].mea`

In [27]: `print(coef)`

```
0.6424814977277227
```

In [28]: `income_df['Annual_HH_Income'].skew`

Out[28]: <bound method Series.skew of 0 64200

1	79920
2	112800
3	97200
4	147000
5	196560
6	167400
7	216000
8	218880
9	220800
10	278400
11	279840
12	292032
13	316800
14	244800
15	246000
16	261000
17	258000
18	348000
19	385200
20	351360
21	445440
22	330480
23	469200
24	466200
25	449400
26	556920
27	412800
28	488880
29	619200
30	523800
31	507600
32	437400
33	610200
34	596160
35	456840
36	570000
37	581760
38	600600
39	590400
40	590400
41	647400
42	756000
43	1075200
44	1142400
45	885600
46	1152480
47	1404000
48	1032000
49	1320000

Name: Annual_HH_Income, dtype: int64>

In [29]: `print(income_df['Annual_HH_Income'].kurtosis())`

1.1012908548129197

In []: